Real-Time Vehicle Detection using YOLOv8 on Android

Technology Stack: YOLOv8, Python, TensorFlow Lite, Android Studio, Java

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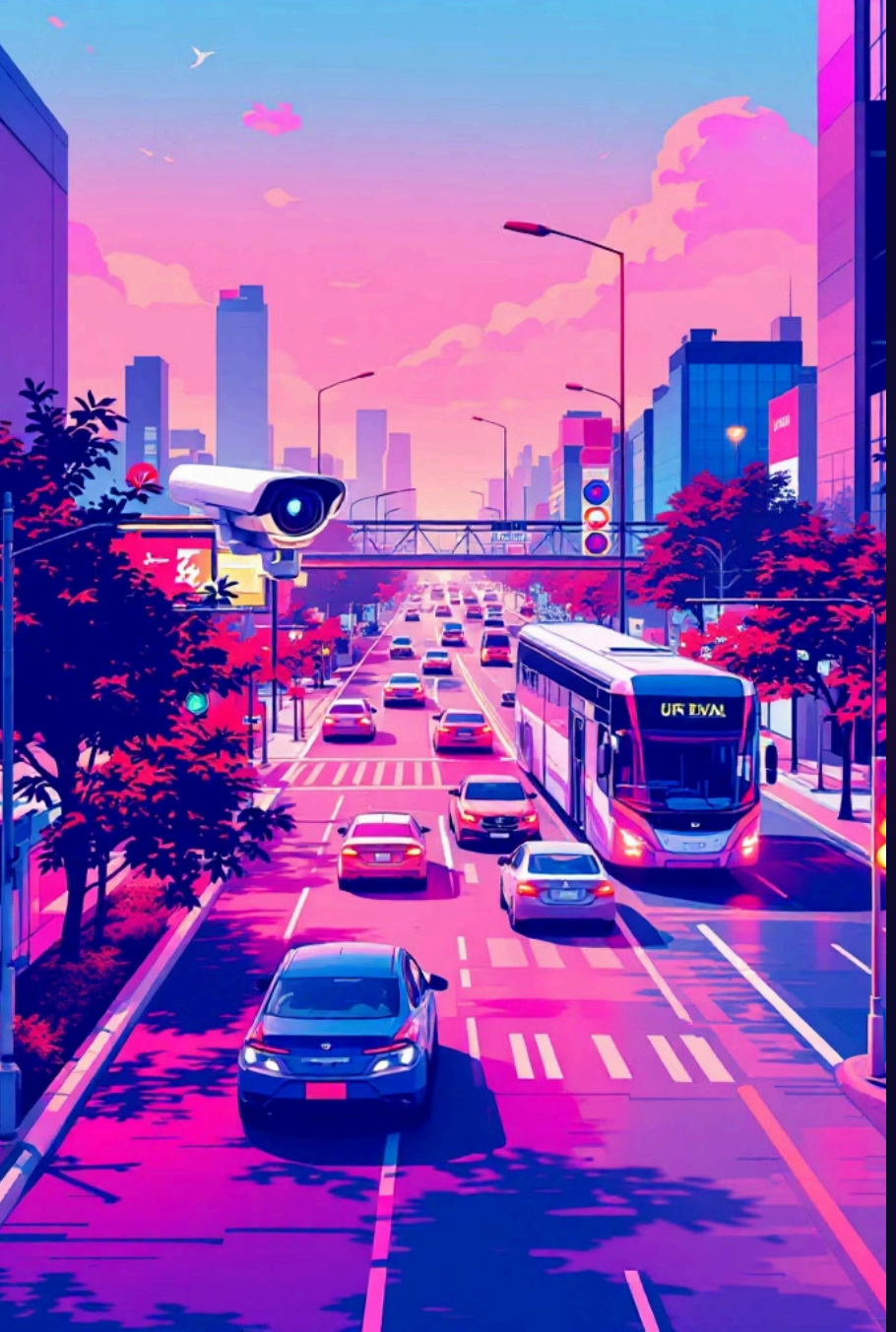

Problem Statement

Real-Time Challenge

Traditional vehicle monitoring systems struggle with latency and processing speed requirements

Mobile Limitations

Need for efficient detection on resource-constrained mobile devices without cloud dependency

Practical Applications

Traffic monitoring, smart city infrastructure, and intelligent surveillance systems

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Our Solution

YOLOv8 Model

Built real-time vehicle detection system using state-of-the-art object detection architecture

Custom Training

Trained on extensive dataset including cars, buses, trucks, and motorcycles

Mobile Deployment

Deployed via TensorFlow Lite for on-device detection without internet

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System Architecture

Dataset Preparation

Collected and annotated images of various vehicle types

Model Training

Trained YOLOv8 architecture using Python and Ultralytics framework

Model Export

Converted .pt model to .tflite format for mobile optimization

Android Integration

Built Java application with CameraX for real-time camera input

Detection Output

Processed frames with bounding boxes and confidence scores

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Technology Stack

AI & Machine Learning

- **YOLOv8** (Ultralytics) for real-time object detection
- **Python** for training pipeline
- **TensorFlow Lite** for mobile deployment

Mobile Development

- **Android Studio** with Java
- **CameraX** for camera integration
- **On-device processing** without internet

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Results & Performance

Multi-Class Detection

Accurately detects cars, trucks, buses, and motorcycles with class labels

Real-Time Processing

Processes live camera feed with bounding boxes and confidence scores

Lightweight Model

Optimized for CPU execution on mobile devices with limited resources

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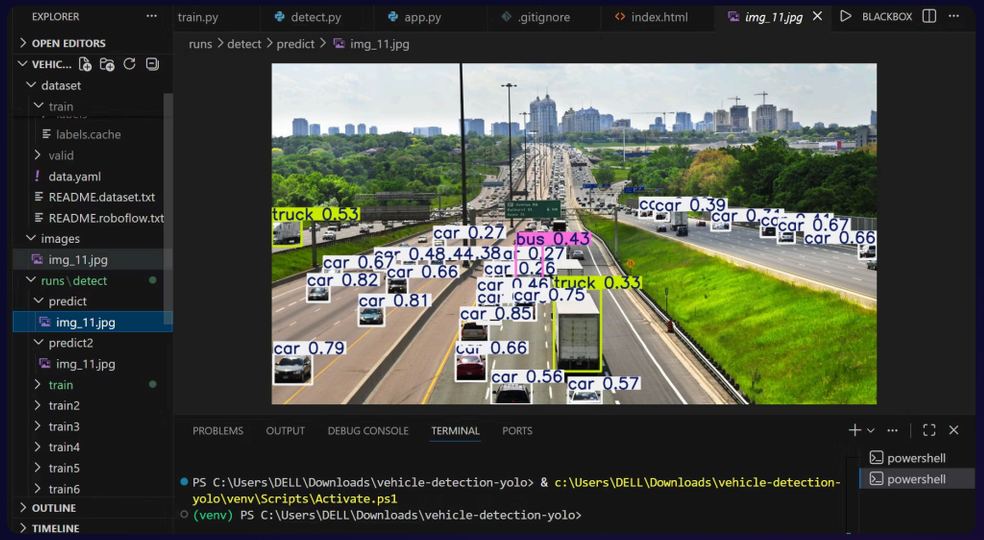

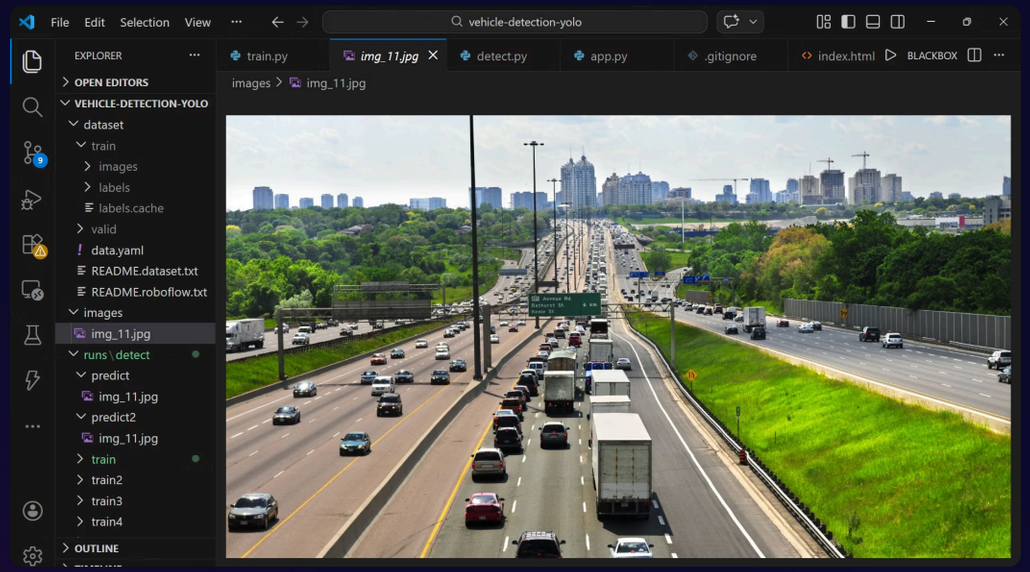

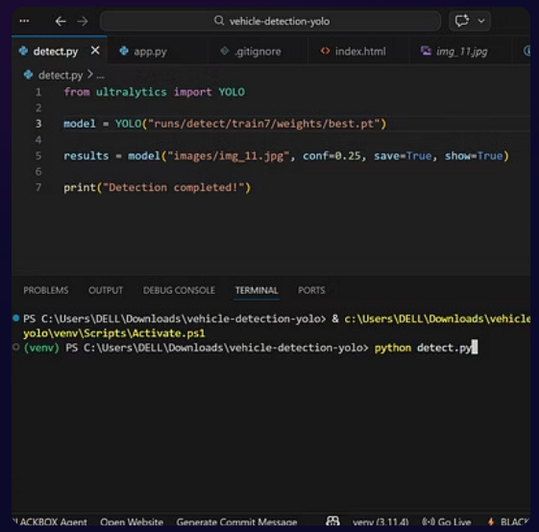

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